

Spoken Tutorial – Introduction to Tokens and Tokenization

Title: Introduction to Tokens and Tokenization

Module	Artificial Intelligence Essentials
Episode	6. Understanding Tokens and Tokenization
Learning Objective	At the end of this tutorial learner will be able to 1. Explain what tokens are, how tokenization works, and why tokens are crucial for LLM performance, cost, and efficiency.
Approx. Duration	3-4 min
Outline	• What are Tokens? • How Tokenization Works? • Why Tokens Matter: Context Windows • Why Tokens Matter: Cost • Why Tokens Matter: Chunking
Meta Tags	LLM, tokens, tokenization, tokenizer, context window, RAG, cost, chunking, AI, large language models
Pre-requisite Tutorial	Introduction to Generative AI, Introduction to Prompting

Script

Visual Cue	Narration
No visual cue provided.	Welcome to this Spoken Tutorial on Introduction to Tokens and Tokenization .
No visual cue provided.	In this tutorial, you will learn to explain what tokens are. Understand how tokenization works. And identify why tokens are crucial for LLM performance, cost, and efficiency.
No visual cue provided.	Here I am using a browser on a computer or mobile. We will also use an online tokenizer tool.
No visual cue provided.	To follow this tutorial, you must be connected to the internet. You should also be familiar with Generative AI and Prompting .

Illustration of LEGO blocks forming a sentence, then a screenshot of an online tokenizer showing "The" as one token and "Tokenization" as multiple tokens.	LLMs don't read words like humans do. They process text in small chunks called tokens. Think of tokens as the fundamental building blocks of language for an AI. It's like breaking down a long sentence into individual LEGO blocks. Let's see this with an example using an online tokenizer tool. Open your browser and search for an online tokenizer. Many free tools are available. Type the word "The" into the tokenizer. Notice it often appears as one token. Now type "Tokenization". Observe how it might be split into multiple tokens, like "Token", "iz", and "ation". This shows how an LLM breaks down text into pieces it can understand.
Illustration of a dictionary or a chef chopping vegetables, then a screenshot of an online tokenizer showing "Hello, how are you today?" split into multiple tokens.	The tokenizer acts like a special dictionary for the LLM. It knows how to split raw text into these sub-word tokens. It's like a chef chopping vegetables into specific sizes for a recipe. Let's continue with our tokenizer tool. Type "Hello, how are you today?" See how the tokenizer splits words, punctuation, and spaces into separate tokens. For example, "Hello", ",", " ", "how", " ", "are", " ", "you", " ", "today", "?". This process, called tokenization, is how an LLM prepares text for understanding.
Illustration of a whiteboard with limited space, then a screenshot of an online tokenizer showing a long paragraph and its token count.	Now, why should you care about tokens? First, consider the LLM's context window. Think of the context window as an LLM's short-term memory. It's like a small whiteboard where the LLM can only see a limited number of words at once. This memory is measured in a fixed number of tokens, for example, 8,192 tokens. Go back to your tokenizer tool. Paste a long paragraph, perhaps from a news article. Observe the token count displayed by the tool. If your text exceeds the LLM's context window limit, it simply gets cut off. This means the LLM cannot "remember" or process everything you give it beyond that token limit.

Illustration of a taxi meter, then a screenshot of an online tokenizer showing token counts for a short sentence versus a long paragraph.	Second, tokens directly impact the cost of using LLMs. API providers charge you per token processed. It's like a taxi meter that charges you for every mile you travel. More miles, more cost. Let's use our tokenizer again. Type a short sentence, then a very long paragraph. Notice the difference in token count. A system like RAG, or Retrieval Augmented Generation, aims to be more efficient. It can reduce the number of tokens sent to the LLM. Fewer tokens mean significantly cheaper operations, especially at scale.
Illustration of a book being divided into chapters, then a visual representation of a document being chunked into token-measured segments.	Finally, tokens are vital for chunking your documents. When preparing large documents for an LLM, we often break them into smaller pieces called chunks. Think of chunking a book into individual chapters or sections. Each chunk needs to fit within the LLM's context window. Imagine you have a large PDF document. Before sending it to an LLM, you would split it into smaller chunks. Each of these chunks must be measured in tokens. This ensures that the LLM can process each part without exceeding its memory limit. Understanding tokens helps you manage and optimize your data effectively.
No visual cue provided.	With this, we come to the end of this tutorial. Let us summarise. We learned that LLMs understand text via tokens. Tokenization is the process of splitting text into these sub-word units. And tokens are critical for context windows, cost, and efficient chunking.
No visual cue provided.	As an assignment, please do the following. Find an online tokenizer tool. Experiment with different types of text: short words, long words, sentences, and paragraphs. Observe how the token count changes. Try to find a sentence that results in exactly 10 tokens.
No visual cue provided.	This Spoken Tutorial is brought to you by EduPyramids Educational Services Private Limited, SINE, IIT Bombay. Thank you for joining!